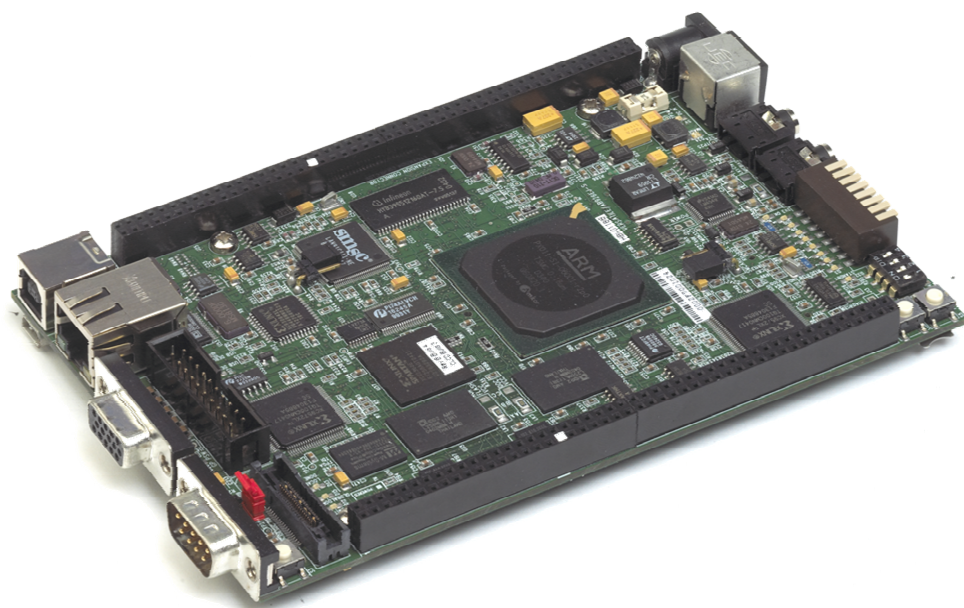


The ARM® RealView® Versatile family of development boards provides an ideal starting point for porting code and developing applications with an ARM processor. This datasheet describes a small form factor application development board that supports both the ARM926EJ-S™ processor and other high performance IP. To expand the baseboard there is a growing range of interface boards available that expand on, and add features to, the basic system.

### Application Baseboard for ARM926EJ-S

The Application Baseboard for the ARM926EJ-S processor is a silicon-vendor independent software development board for the ARM926EJ-S and PrimeXsys™ developer community. It is also suitable for advanced 3D graphics application development around ARM and PowerVR MBX processor.

At the heart of the board is a highly integrated development chip which is based on the ARM926EJ-S PrimeXsys Platform architecture. External bus interfaces on the development chip, which enable bus layers inside the chip to be routed off-chip, allow additional peripherals to be added in an FPGA on the baseboard.



The development chip is also used in the RealView® Platform Baseboard for ARM926EJ-S, a fully featured ASIC emulation and prototyping board. Because both boards are based on the same device, with the same memory and peripheral map, upgrading is easy. Moreover OS ports and applications will migrate seamlessly between platforms.

#### Features of the ARM926EJ-S Development Chip

- ARM926EJ-S processor, Jazelle® Java acceleration hardware and DSP extensions
- PowerVR MBX 3D Graphics accelerator
- MOVE™ coprocessor, hardware acceleration for MPEG encode
- Vector Floating Point (VFP) coprocessor, hardware assist for floating point arithmetic
- Static and dynamic memory sub-system including DMA controller
- Vectored interrupt controller
- Peripherals essential for booting operating systems

#### Features of the Application Baseboard

- Display, audio and networking capability built-in as standard
- Support for JTAG run-control and Trace, including a built-in USB debug port
- Small form factor 160x100mm
- Integrated multi-region power supply
- Easy expansion through 0.1" pitch headers
- Migration path to full ASIC prototyping environment, which has PCI and AHB expansion

## Application Baseboard for ARM926EJ-S Specification

### Performance

- CPU to 210MHz
- Memory, graphics and internal bus to 75MHz
- FPGA to 35MHz

### Memory

- 64MB NOR Flash
- 64MB Disk-on-chip NAND Flash
- 128MB 32-bit SDRAM and 2MB SRAM

### Peripherals

- Ethernet controller
- Color LCD controller with VGA monitor output
- Serial port (UART0)
- TD243 USB 2.0 OTG (host and device) controller
- SmartCard (SIM) reader
- Keyboard and mouse controllers
- MMC/SD Card reader
- Stereo audio codec (including mic input)
- Vectored interrupt controller
- Built-in JTAG run-control hardware

### Expansion signals (2 x 96-way 0.1" headers)

- Static memory interface
- LCD and touchscreen interface
- 2 Serial ports (UART1/UART2)
- Synchronous Serial Port (SSP)
- 16 GPIO signals
- 2 interrupt inputs

### Design support

- Verilog RTL for baseboard FPGA and PLDs

### Documentation

- Application Baseboard for ARM926EJ-S User Guide
- ARM926EJ-S Development Chip Reference Manual
- PrimeCell® Peripheral Technical Reference Manuals

### Firmware

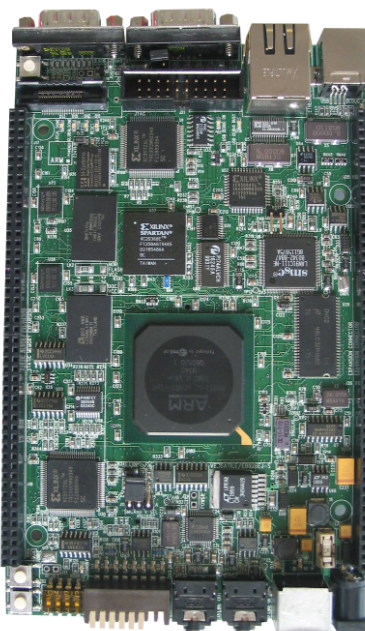
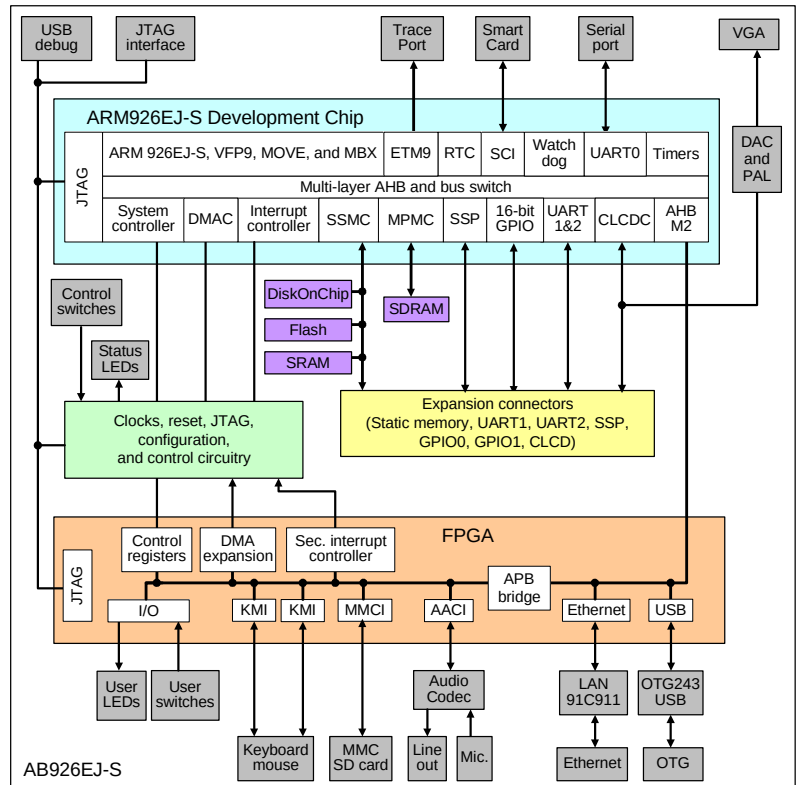
- System and memory initialization code
- Polled serial drivers
- Real-time clock (time function) in retargeted C library
- Disk-on-Chip, NAND flash file system
- NOR Flash memory read, write and erase code
- Serial and Ethernet image transfer to memory
- Vectored interrupt controller example
- Timer example
- Hardware vector floating point (VFP) example
- Functional test program for all peripherals
- C library support for stand-alone and semi-hosted images
- Simple makefiles
- Little and big-endian build options

### Pre-built binaries

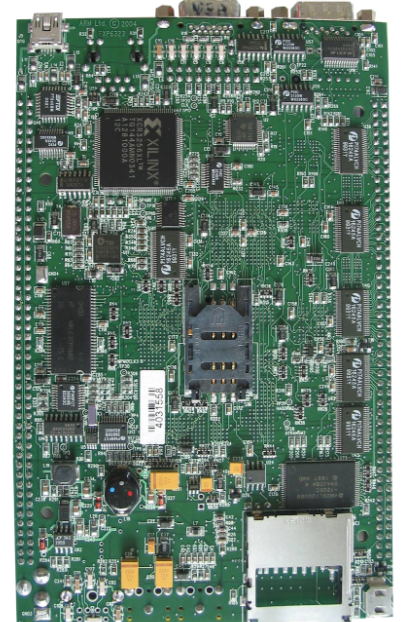
- Boot monitor
- Functional test program
- ARM Embedded Linux

### Ordering Information

| Part number    | Description                                   |
|----------------|-----------------------------------------------|
| AB926-BD-0221A | RealView Application Baseboard for ARM926EJ-S |



Front View



Back View

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